

CS1CA Computer Systems Architecture

Memory hierarchy Design (2)

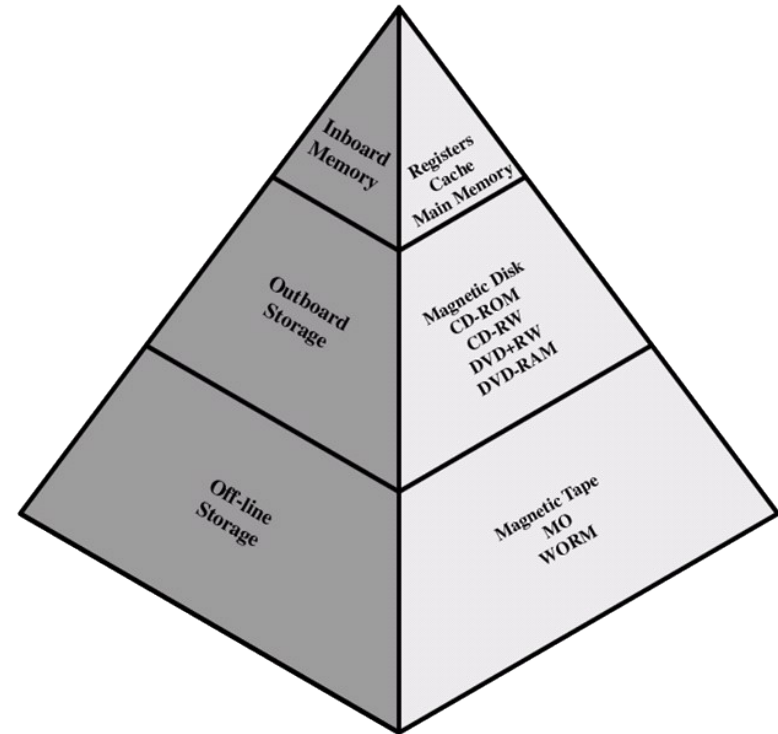
Week 10 lecture 2 of autumn semester 2025

Lectured by

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Outline of the topic

- General introduction to memory hierarchy
- Cache and cache performance measure
 - Techniques for reducing cache miss penalty
 - Techniques for reducing cache misses
 - Techniques for reducing hit time
- Main memory: different organisations
- Virtual memory
- Self-learning: Memory hierarchy design in the Alpha 21264

In this lecture

- The following contents will be delivered
 - Cache performance measure
 - Techniques for reducing cache miss penalty
- The learning outcomes are listed below
 - To be able to use “average memory access time (AMAT)” to measure cache performance
 - To derive the AMAT equation for cache performance evaluation
 - To be able to use introduced techniques to reduce cache miss penalty

Cache performance measure

Average memory accesstime (AMAT)

- ***Hit time***: time to hit in the cache (unit: nanosecond or number of clock cycles);
 - ***Miss rate***: the fraction of accesses that are not in the cache;
 - ***Miss penalty***: the additional cycles to service the miss (unit: nanosecond or number of clock cycles).
- **Derivation of the cache performance equation.**

Effect of cache on CPU time - 1

$$\text{CPU time} = (\text{CPU clock cycles} + \text{memory stall cycles}) \times \text{clock cycle time}$$

In the above equation:

$$\text{CPU clock cycles} = IC \times CPI$$

Assumptions

- All memory stall cycles are due to cache misses;
- Cache hit is part of the CPU execution cycle;
- read/write miss penalties are the same (not necessarily true)

$$\text{Memory stall cycles}$$

$$= \text{No. of misses} \times \text{Miss penalty}$$

$$= IC \times \text{Misses per instruction} \times \text{Miss penalty}$$

$$= IC \times \text{Memory references per instruction} \times \text{Miss rate} \times \text{Miss penalty}$$

Effect of cache on CPU time - 2

- Substituting CPU clock cycles and memory stall cycles to the CPU time equation, we have

CPUtime

$$= [(IC \times CPI) + (IC \times \text{Memory references per instruction} \times \text{Miss rate} \times \text{Miss penalty})] \times \text{Clock cycle time}$$

$$= IC \times [CPI + (\text{Memory references per instruction} \times \text{Miss rate} \times \text{Miss penalty})] \times \text{Clock cycle time}$$

Example:

Given data: Cache miss penalty = 50 clock cycles
CPI = 2 clock cycles (no cache misses)
Miss rate = 2% (depends on cache design)
Memory references per instruction = 1.33

$$\begin{aligned} CPU\ time_{without\ cache} &= IC \times (2 + 1.33 \times 100\% \times 50) \times Clock\ cycle\ time \\ &= IC \times 68.5 \times Clock\ cycle\ time \end{aligned}$$

$$\begin{aligned} CPU\ time_{with\ cache} &= IC \times (2 + 1.33 \times 2\% \times 50) \times Clock\ cycle\ time \\ &= IC \times 3.33 \times Clock\ cycle\ time \end{aligned}$$

$$CPU\ time_{perfect\ cache} = IC \times 2 \times Clock\ cycle\ time$$

❖ The inclusion of a cache has a significant effect on CPU execution time.

To improve cache performance

- The measure of cache performance is **Average Memory Access Time**, which is represented by three components as

Average memory accesstime

- To improve cache performance, we can
 - reduce the miss rate;
 - reduce the miss penalty;
 - reduce the hit time.

Reducing miss penalty 1: multilevel caches

- Add another level of cache between the original cache and RAM;
- The first level cache (close to CPU) can be small and fast enough to match the clock cycle time of the CPU.
Provide fast cache access;
- The second level cache (close to RAM) can be larger so as to capture many accesses that would go to RAM.
Reduces the effective miss penalty;
- The concept is straightforward; but the performance analysis is slightly complicated.

Reducing miss penalty 1: multilevel caches

- Definition of average memory access time for a two-level cache.

Average memory access time

$$= \text{hit time}_{L1} + \text{miss rate}_{L1} \times \text{miss penalty}_{L1}$$

and

miss penalty_{L1}

$$= \text{hit time}_{L2} + \text{miss rate}_{L2} \times \text{miss penalty}_{L2}$$

where subscripts L1 and L2 refer to a first-level and a second level cache, respectively.

Reducing miss penalty 1 : multilevel caches

$$\text{Average memory access time} = \text{hit time}_{L1} + \text{miss rate}_{L1} \times (\text{hit time}_{L2} + \text{miss rate}_{L2} \times \text{miss penalty}_{L2})$$

- **Local miss rate:** no. of misses in the cache divided by the total no. of memory accesses to this cache.
- **Global miss rate:** no. of misses in the cache divided by the total no. of memory accesses generated by the CPU; in the above, the globe miss rate of the two-level cache is

$$\text{Miss rate}_{L1} \times \text{Miss rate}_{L2}$$

Reducing miss penalty 1: multilevel caches

Example

- 1000 memory references;
- 40 misses in the first-level cache;
- 20 misses in the second-level cache;
- What are the various miss rates?

Miss rate for the first-level cache: $40/1000 = 4\%$

Miss rate for the second-level cache: $20/40 = 50\%$

Global miss rate is

$$20/1000 = 2\% \text{ or } 4\% \times 50\% = 2\%$$

Reducing miss penalty 2: early restart and critical word first

- Based on the observation that the CPU needs only one word of the block at a time.
- Two specific strategies:
 - Early restart: as soon as requested word arrives, send to the CPU to continue the process;
 - Critical word first: request a word from memory & send the word to processor, processor continues and the block is filled with the rest of the words.
- This technique only benefits cache designs with very large cache blocks.

Reducing miss penalty 3: victim caches

- Victim cache is a small, fully associative cache between a cache and RAM;
- This victim cache contains blocks that are discarded from a cache because of a miss.
- On a cache miss, the victim cache is checked first to see if the required data is there before going to the next lower-level memory.
- If the required data is found in the victim cache, the victim block and cache block are swapped.

End of this lecture

Questions and exercises – 1

- What is the measure of cache performance?
- Can you derive the equation which measures cache performance?
- Explain the three techniques for reducing miss penalty.

End of this lecture

Questions and exercises – 2

- Adding clock cycles used for memory access into account, calculate the CPU time with the following given data.
 - Cache miss penalty = 60 clock cycles
 - CPI = 2 clock cycles (without considering memory access)
 - Miss rate = 3%
 - Memory references per instruction = 1.24

If reducing cache miss penalty to 30 clock cycles and keeping other data the same, calculate the CPU time again.
- For a two-level cache, there are 150 memory references from instructions. Among them, 30 cache misses are at the first-level cache and 3 misses at the second-level cache. Calculate the local miss rate at each level and the global miss rate.